Scale-Adaptive Document Conditioning and Gated CNN–Attention–BiLSTM Recognition for Multi-Dataset Offline Handwritten Text Recognition

A. Venkata Balaji

***Abstract*—Offline handwritten text recognition (HTR) requires a model to interpret character sequences from document images whose writing style, scale, background, and layout can vary considerably. These variations are particularly relevant when historical manuscripts, handwritten numerical strings, and contemporary handwriting are evaluated within the same study. This paper presents a compact HTR pipeline that couples scale-adaptive document conditioning with a hybrid Gated Convolutional Neural Network (CNN), Multi-Head Self-Attention (MHSA), Bidirectional Long Short-Term Memory (BiLSTM), and Connectionist Temporal Classification (CTC) recognizer. The conditioning stage estimates a document-specific median component height and uses it to parameterize selected image-processing operations, including illumination normalization, deskewing, ruled-line suppression, text-region extraction, and geometric filtering. The recognition network contains seven convolutional stages, six residual gated-convolution blocks, a two-head self-attention layer, three bidirectional recurrent layers, and a dataset-specific character projection. For the IAM Combined vocabulary of 83 output classes, the stored model contains 1,583,847 parameters. We report dataset-specific experiments on Parzival, Saint Gall, ORAND-CAR-A, CVL Digits, and IAM Combined. Held-out test evaluations yield 2.48% CER and 8.53% WER on Parzival, 8.38% CER and 41.62% WER on Saint Gall, 3.33% CER on ORAND-CAR-A, and 50.78% CER on CVL Digits. IAM Combined is reported separately with a best validation edit distance of 5.66%; no independent test result is available for this run. The large validation-to-test discrepancy on CVL Digits is treated as a diagnostic finding and not as evidence of cross-domain generalization. The study provides a traceable evaluation of a compact hybrid recognizer and identifies the conditions under which its current results should be interpreted.**

***Index Terms*—Offline handwritten text recognition, document image processing, scale-adaptive conditioning, gated convolution, self-attention, bidirectional LSTM, connectionist temporal classification, character error rate.**

I. Introduction

OFFLINE handwritten text recognition converts a static image of handwriting into a machine-readable sequence. Unlike online recognition, the system does not observe the order or timing of pen movements. It must infer character identity and sequence structure from the final visual trace. The task is therefore affected by writer variation, connected cursive strokes, irregular spacing, slant, degraded paper, and ambiguity between visually similar characters. These factors become more pronounced in historical documents and in application-specific material such as handwritten amounts and digit strings.

Recent HTR systems have explored convolutional-recurrent networks, gated convolutional encoders, self-attention, Transformer-based recognition, and end-to-end document models. Gated fully convolutional recognition offers an alternative to recurrent encoders for line-level transcription [1]. Comparative work on text-line recognition has also shown that the choice of encoder and decoder is consequential: self-attention coupled with CTC is a viable alternative to an autoregressive Transformer decoder under suitable training and language-model conditions [2]. More recent systems use pretrained visual and language representations, including TrOCR and decoder-only recognition architectures [3], [4]. At document level, attention networks can integrate segmentation and recognition rather than relying on a separately extracted line sequence [5], [6].

These approaches address different recognition settings and are not directly interchangeable. A page-level model, a line recognizer, and a digit-string recognizer operate on different input units and often use different training resources. Likewise, a pretrained model evaluated after large-scale external pretraining cannot be compared with a compact model trained on a target corpus without disclosing the data and initialization differences. Dataset and protocol variation is a recurring issue in HTR benchmarking, particularly for historical collections with different scripts, writers, annotation units, and partitioning conventions [10], [11].

This study examines a practical alternative: a compact line-recognition pipeline in which document conditioning is performed before a gated convolutional and recurrent recognizer. The conditioning stage uses a document-specific estimate of median connected-component height to scale selected morphological and geometric operations. The recognizer then combines local convolutional features, self-attention, bidirectional sequence modelling, and CTC decoding. The intention is not to claim a new family of attention or CTC operation. Rather, the contribution is the implementation and evaluation of this integrated pipeline across several distinct handwriting corpora, with explicit reporting of where test evidence is and is not available.

The study addresses the following questions:

- **RQ1:** How can document-specific scale estimates parameterize a sequence of conditioning operations before line recognition?

A. Venkata Balaji is with the School of Computer Science and Engineering, Vellore Institute of Technology (VIT), Vellore, India (e-mail: galigastagiramma@gmail.com).

The implementation and supplementary experimental artifacts are available at: https://github.com/Balaji268268/HandwrittenTextRecognition.

- **RQ2:** What recognition architecture is implemented by combining residual gated convolution, self-attention, and bidirectional recurrent modelling?
- **RQ3:** How does the resulting system perform on the available held-out test partitions of heterogeneous handwriting datasets?
- **RQ4:** What does the difference between validation and test performance on CVL Digits indicate about the limits of the current evaluation?

The contributions are fourfold. First, we describe a scale-adaptive conditioning pipeline in which selected image-processing parameters are derived from the median height of eligible connected components. Second, we document the implemented Gated CNN–MHSA–BiLSTM–CTC architecture at the level of its actual stages and dimensions. Third, we report dataset-specific training, validation, and held-out test evidence without treating unlike evaluation levels as equivalent. Fourth, we discuss the substantial CVL Digits validation-to-test discrepancy and the limitations that prevent broader generalization claims.

II. Related Work

A. Convolutional, Recurrent, and CTC-Based Recognition

CNN–RNN–CTC systems remain an important line-recognition design because convolutional layers extract local visual patterns while recurrent layers model dependencies across the horizontal sequence. CTC removes the need to provide character-level alignment during training by summing over valid frame-level alignments [12]. Gated convolutional networks have been studied as a way to modulate visual features and, in fully convolutional variants, reduce dependence on recurrent computation [1], [13]. The present model retains recurrent sequence modelling, but uses residual gated convolutional blocks within its visual encoder.

Hernandez Diaz *et al.* compared several encoder and decoder combinations for text-line recognition, including BiLSTM, self-attention, gated recurrent convolutional layers, CTC, and Transformer decoding [2]. Their analysis motivates treating encoder and decoder choices as separable design decisions. In our implementation, self-attention is applied to the convolutional feature sequence before three bidirectional recurrent layers, while CTC supplies the alignment-free training objective.

B. Attention and Transformer-Based HTR

Attention-based methods have expanded HTR beyond conventional recurrent encoders. TrOCR uses pretrained image and text Transformers and demonstrates the value of large-scale pretraining for recognition [3]. DTrOCR explores a decoder-only Transformer formulation for printed, handwritten, and scene text [4]. HTR-VT investigates a data-efficient Vision Transformer encoder for handwriting recognition [7]. At the paragraph and document levels, the Vertical Attention Network and Document Attention Network integrate text localization and recognition through learned attention mechanisms [5], [6].

These methods differ in recognition unit, training data, decoder design, and use of external pretraining. Consequently, a reported paragraph-level CER or a pretrained Transformer result should not be interpreted as a directly matched baseline for the line-level, dataset-specific experiments in this paper. The comparison in Section V therefore reports the protocol context alongside published results and avoids ranking unlike experiments as if they shared a common test setting.

C. Compact Recognition and Data Efficiency

Compact HTR remains relevant when the training corpus is limited or the target application constrains model size. Recent work has examined recursive architectures for compact HTR and self-supervised learning to reduce reliance on labelled handwriting data [8], [9]. These directions are complementary to the present study. The current implementation uses supervised dataset-specific training for most reported evaluations and a warm-started IAM Combined run; it does not include the large-scale self-supervised pretraining used by LoGo-HTR.

D. Document Conditioning and Dataset Protocols

Document preprocessing and line segmentation remain consequential because recognition models are sensitive to the quality and consistency of their input regions. Historical HTR datasets also differ in script, writer composition, page layout, annotation level, and partition design. Surveys of HTR datasets document this heterogeneity and reinforce the need to state the corpus and evaluation protocol precisely [10], [11]. The conditioning rules used here are deterministic image-processing heuristics. Their presence in the pipeline does not, by itself, establish that they improve CER; a controlled preprocessing ablation would be required for that causal claim.

III. Problem Formulation and Proposed Method

A. Recognition Task

Let an input line image be denoted by $I \in \mathbb{R}^{H \times W}$ and its transcription by $y = (y_1, \ldots, y_U)$, where each y_u belongs to a dataset-specific character vocabulary $\mathcal{V}$. The recognition model produces a sequence of frame-level distributions over $\mathcal{V}$ augmented with the CTC blank symbol. The task is to infer a transcription from the image without requiring character bounding boxes or character-level alignment.

The full processing path is represented as

$$\hat{y} = \mathcal{D}_{\mathrm{CTC}}(\mathcal{R}_\theta\left(\mathcal{P}(I)\right)), \tag{1}$$

where $\mathcal{P}$ is the document-conditioning and line-extraction procedure, $\mathcal{R}_\theta$ is the neural recognizer, and $\mathcal{D}_{\mathrm{CTC}}$ is the beam-search decoder.

B. Scale-Adaptive Document Conditioning

The conditioning module first converts the input to grayscale and determines foreground polarity. It then estimates a characteristic document scale from connected components. Given component heights $\{h_i\}_{i=1}^{N}$ that satisfy the implementation's eligibility constraints, the scale estimate is

$$h_m = \mathrm{median}\{h_i : i \in \mathcal{E}\}, \tag{2}$$

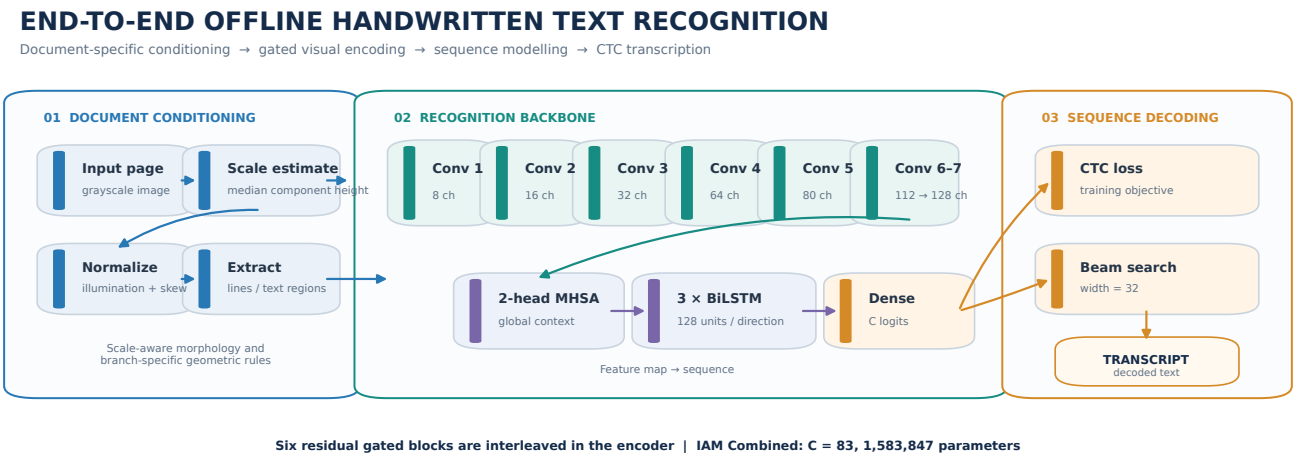


Fig. 1. System-level workflow of the proposed offline HTR pipeline. Document conditioning produces line or region images for the recognition network. The recognition network comprises seven convolutional stages, six residual gated-convolution blocks, two-head MHSA, three BiLSTM layers, and a dataset-dependent character projection. CTC beam search with width 32 is used for decoding.

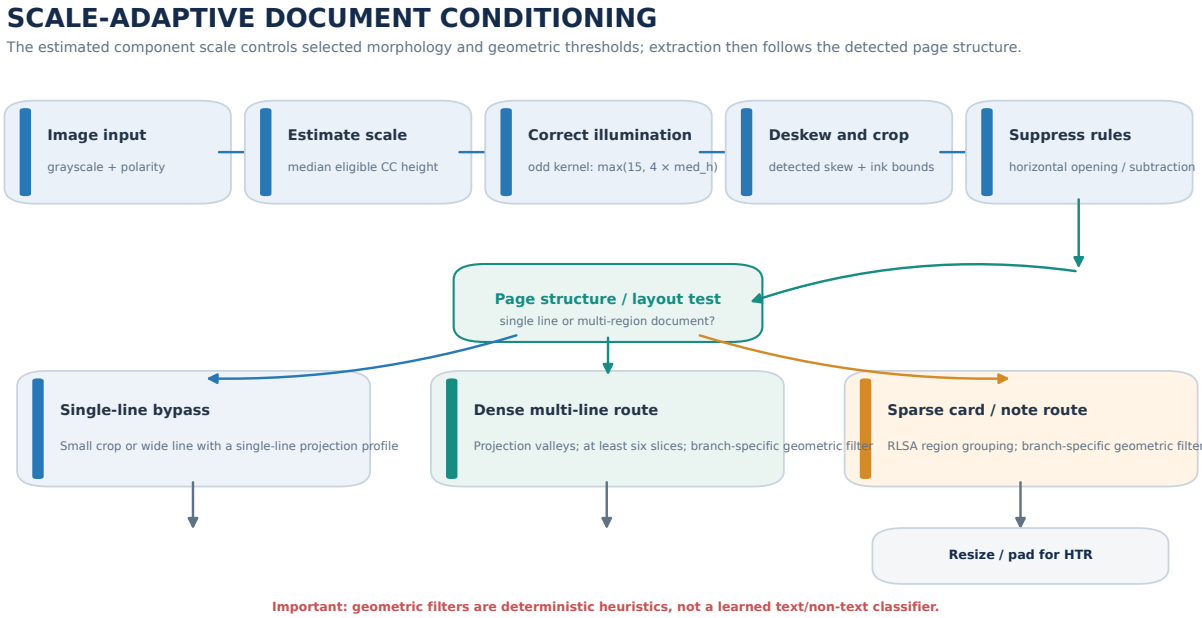


Fig. 2. Scale-adaptive document conditioning. The median connected-component height is used to parameterize selected operations. The extraction stage includes a single-line bypass and separate dense multi-line and sparse card/note branches; the geometric rejection rules are branch-specific heuristics rather than a universal non-text classifier.

where $\mathcal{E}$ is the set of eligible components. When fewer than three components satisfy the constraints, the implementation uses a fallback based on document height. This estimate is an operational scale statistic; it is not a direct measurement of physical DPI or true character height.

For illumination normalization, the implementation constructs a rectangular morphological-closing kernel whose side length is the next odd integer at or above $\max(15, 4h_m)$. If $B(I)$ denotes the estimated background, the normalized image is computed by a clipped background-division operation:

$$I_{\text{norm}}(x,y) = \text{clip}\left(\frac{I(x,y)}{\max(B(I)(x,y),1)}\,255, 0, 255\right). \quad (3)$$

Deskewing estimates the dominant foreground orientation and applies a rotation when the detected angle lies within the implementation's accepted range. The image is then cropped around the foreground with padding related to h_m. Ruled-line suppression uses horizontal morphological opening followed by subtraction from the foreground mask.

Text extraction first checks a single-line bypass condition; if the cropped region is sufficiently short or wide and the smoothed projection indicates one line, the image is passed through without multi-line segmentation. Otherwise, extraction follows two conditional paths. In dense documents, a smoothed horizontal projection profile is used to identify candidate valleys.

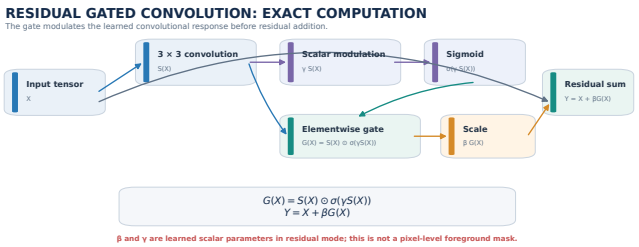


Fig. 3. Residual GatedConv2D operation reconstructed from the implementation. The convolutional response is modulated by a sigmoid term controlled by the learned scalar γ, scaled by β, and added to the input tensor.

When at least six vertical slices are detected, candidate slices are checked for minimum content and filtered using a branch-specific aspect-ratio, height, and vertical-position condition. In sparse card or note regions, horizontal dilation and connected-region grouping are used to form candidates; the second branch applies its own aspect-ratio, height, and vertical-position rule. The implementation therefore contains two related but non-identical geometric heuristics. Neither should be described as a learned doodle classifier, and the current experiments do not quantify false removal of genuine handwriting.

C. Gated Convolutional Feature Extraction

The visual encoder contains seven convolutional stages with channel widths $8, 16, 32, 64, 80, 112$, and 128. Each stage uses a 3×3 convolution, batch normalization, and a Swish activation. Six residual GatedConv2D blocks are interleaved with the convolutional stages. The pooling schedule reduces the spatial dimensions while retaining a horizontal sequence suitable for line recognition.

For a residual gated block, let X denote its input and let $S(X)$ be the convolutional projection. The implementation computes

$$G(X) = S(X) \odot \sigma(\gamma S(X)), \tag{4}$$

followed by

$$Y = X + \beta G(X), \tag{5}$$

where β and γ are learned scalar parameters in the residual mode. This is the implemented residual gating formulation. It modulates the convolutional response before adding it to the input; it should not be interpreted as a separately trained pixel-level foreground mask.

D. Self-Attention and Bidirectional Sequence Modelling

The encoder output is reshaped into a sequence of feature vectors. A Multi-Head Attention layer is applied to this sequence with two heads and a key dimension of eight per head for the 128-channel encoder output. The attention operation is

$$\mathrm{MHA}(Q, K, V) = \mathrm{Concat}(H_1, H_2)W^O, \tag{6}$$

where each head uses scaled dot-product attention. The attended sequence is then passed through three bidirectional LSTM stages, each with 128 units in the forward and backward directions. The two directional outputs are concatenated at each time step.

A dense projection maps the resulting sequence representation to C dataset-specific output classes. The value of C depends on the tokenizer associated with the checkpoint. For the IAM Combined model, $C = 83$.

E. CTC Objective and Beam-Search Decoding

CTC defines a many-to-one mapping $\mathcal{B}$ from frame-level paths to output strings by removing blanks and collapsing repeated labels. For target sequence y and frame-level probabilities $p(\pi_t \mid I)$, the CTC objective is

$$P(y \mid I) = \sum_{\pi \in \mathcal{B}^{-1}(y)} \prod_{t=1}^{T} p(\pi_t \mid I), \tag{7}$$

with training loss $\mathcal{L}_{\mathrm{CTC}} = -\log P(y \mid I)$. At evaluation, the implementation uses TensorFlow CTC beam search with beam width $W = 32$, one returned path, repeated-label merging, and the configured blank index. The decoder exponentiates the returned log score. This value is described here as an uncalibrated beam-search-derived sequence likelihood. It is not a calibrated confidence estimate, and finite-width beam search should not be described as exhaustive enumeration of every valid alignment path.

IV. Experimental Setup

A. Datasets and Partitions

The experiments cover five corpora with different handwriting characteristics and output vocabularies. The IAM and CVL collections are established offline handwriting resources with distinct writer and transcription characteristics [14], [15]. The test counts and partition counts below are taken from the frozen notebook execution outputs used for the reported evaluations. Dataset-specific training is used for Parzival, Saint Gall, ORAND-CAR-A, and CVL Digits. IAM Combined is a separate warm-started experiment and is reported using validation evidence only.

B. Training Protocol

1) IAM Combined warm-started experiment: The archived training histories contain ten epochs for Parzival, 35 for CVL Digits, 35 for ORAND-CAR-A, and 15 for IAM Combined. The IAM Combined run is a separate warm-started experiment, not a fresh-initialization run. It loads the previously trained IAM checkpoint (`model_iam.weights.h5`) and continues training with the combined IAM configuration. The tokenizer defines 83 output classes, and the resulting architecture contains 1,583,847 parameters. Its best recorded validation edit distance is 5.66% at epoch 12; the final recorded epoch has a higher validation distance. The available artifacts do not provide an independent held-out IAM Combined test evaluation. Therefore, 5.66% is a validation result and must not be reported as IAM test CER or used as a direct test-set comparison with published IAM benchmarks. The Washington checkpoint is retained as a historical baseline artifact with 15 training epochs but without a separate validation metric in its archived history. It is not included in the four held-out test results.

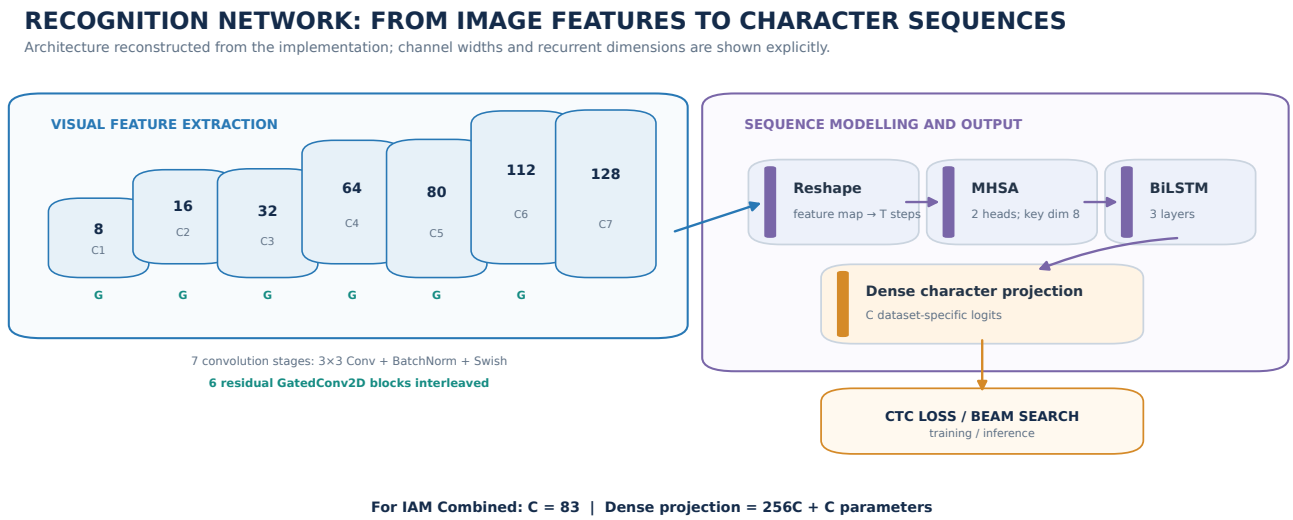


Fig. 4. Implemented recognition architecture. Seven convolutional stages are interleaved with six residual GatedConv2D blocks. The convolutional feature map is reshaped into a sequence, processed by two-head MHSA and three BiLSTM layers, and projected to dataset-specific character logits.

TABLE I
DATASET-SPECIFIC PARTITION AND EVALUATION EVIDENCE. THE IAM COMBINED TEST RESULT IS UNREPORTED.

Dataset	Train	Validation	Test	C	Test evidence	Result
Parzival	2,237	912	1,328	56	Notebook evaluation	CER 2.48%, WER 8.53%
Saint Gall	468	235	707	49†	Notebook evaluation	CER 8.38%, WER 41.62%
ORAND-CAR-A	1,708	301	3,784	14	Notebook evaluation	CER 3.33%
CVL Digits	1,073	189	6,698	14	Notebook evaluation	CER 50.78%
IAM Combined	Not reported	966	–	83	Validation only	Best edit distance 5.66%

† Saint Gall class count is reported in the notebook context; its exported checkpoint and tokenizer are unavailable.

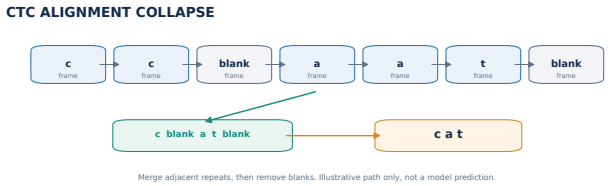


Fig. 5. Illustration of the CTC collapse mapping. Consecutive repeated labels are merged and blank symbols are removed to obtain the output transcription. The path is illustrative and is not a model prediction.

The stored model parameter count follows

$$P_{\text{total}} = 1,562,516 + 257C, \tag{8}$$

where the first term is the shared backbone and the second term accounts for the 256-dimensional dense projection and its bias. For $C = 83$, the IAM Combined checkpoint contains 1,583,847 parameters. This count describes the serialized architecture; it is not an inference-latency measurement.

C. Evaluation Metrics

The evaluator lowercases text, replaces newline characters with spaces, inserts spacing around punctuation, and collapses repeated whitespace. It computes normalized edit distance separately for each sample and averages the per-sample values. Accordingly, the reported CER and WER are macro-averaged per-sample normalized edit distances rather than corpus-level edit rates obtained by pooling all characters or words before division. We retain the notebook's reported metric values and make this aggregation protocol explicit.

EXPERIMENTAL COVERAGE AND EVIDENCE STATUS

Dataset	Train	Validation	Test	Classes	Init.	Evidence
Parzival	2,237	912	1,328	56	Fresh	**Test + logs**
Saint Gall	468	235	707	49*	Fresh	**Notebook test**
ORAND-CAR-A	1,708	301	3,784	14	Fresh	**Test + logs**
CVL Digits	1,073	189	6,698	14	Fresh	**Test + logs**
IAM Combined	Not reported	966	Not reported	83	Warm-start	**Validation only**

* Saint Gall class count is noted in the notebook context; its exported checkpoint and tokenizer are unavailable for independent verification.
IAM Combined is a warm-started run: best validation edit distance 5.66% at epoch 12; no independent held-out test result is present.

Fig. 6. Dataset-specific partition sizes, output dimensions, and evidence availability. The table distinguishes held-out test evaluations from the IAM Combined validation-only experiment.

SELECTED IAM BENCHMARKS REPORTED IN THE LITERATURE

Contextual comparison only: training data, pretraining, decoding, and protocol differ across studies.

Method	Year	CER (%)	WER (%)	Params	Protocol distinction
Self-Attention + CTC + LM	2021	2.75	--	22M	Internal + public data; character n-gram LM
TrOCR-Small	2023	4.22	--	62M	Pretrained visual and text Transformers
TrOCR-Base	2023	3.42	--	334M	Pretrained visual and text Transformers
TrOCR-Large	2023	2.89	--	558M	Pretrained visual and text Transformers
HTR-VT	2025	4.70	14.90	53.5M	Vision Transformer encoder
LoGo-HTR	2026	2.01	10.27	6.4M	External self-supervised handwriting; full IAM labels
This study: IAM Combined	2026	--	--	1.584M	Validation 5.66%; no held-out IAM test

Do not interpret this table as a leaderboard. The present IAM Combined run has validation evidence only, so its test CER/WER are intentionally blank.

Fig. 7. Selected IAM benchmark results reported in published HTR studies. The comparison is contextual rather than a leaderboard because training data, pretraining, language-model use, and evaluation protocols differ. The IAM Combined experiment in this study has validation evidence only, so no test CER or WER is entered.

V. Results and Analysis

A. Training and Validation Behaviour

Fig. 8 plots the training and validation edit-distance histories available in the saved CSV files. The best validation point is marked for each run. The IAM Combined curve must be interpreted in the context of its warm-start initialization. The Washington history is omitted from this plot because its archived CSV does not contain validation metrics. Saint Gall is also omitted because a corresponding epoch-history CSV is not present in the current saved-model inventory.

B. Held-Out Test Performance

The four test-evaluated datasets produce different error rates, reflecting differences in writing style, script, annotation unit, and character distribution. Parzival records 2.48% CER and 8.53% WER. ORAND-CAR-A records 3.33% CER on 3,784 test samples. Saint Gall records 8.38% CER and 41.62% WER. CVL Digits records 50.78% CER on 6,698 test strings. These values are dataset-specific outcomes and do not constitute a single pooled score or a zero-shot transfer experiment.

C. CVL Digits Validation-to-Test Discrepancy

CVL Digits shows a marked difference between its best validation edit distance of 6.93% and held-out test CER of 50.78%, a gap of 43.85 percentage points. This difference warrants investigation. Potential contributors include differences in string-length distributions, writer composition, character frequencies, sample quality, partition construction, and transcript normalization. The available aggregate metrics do not

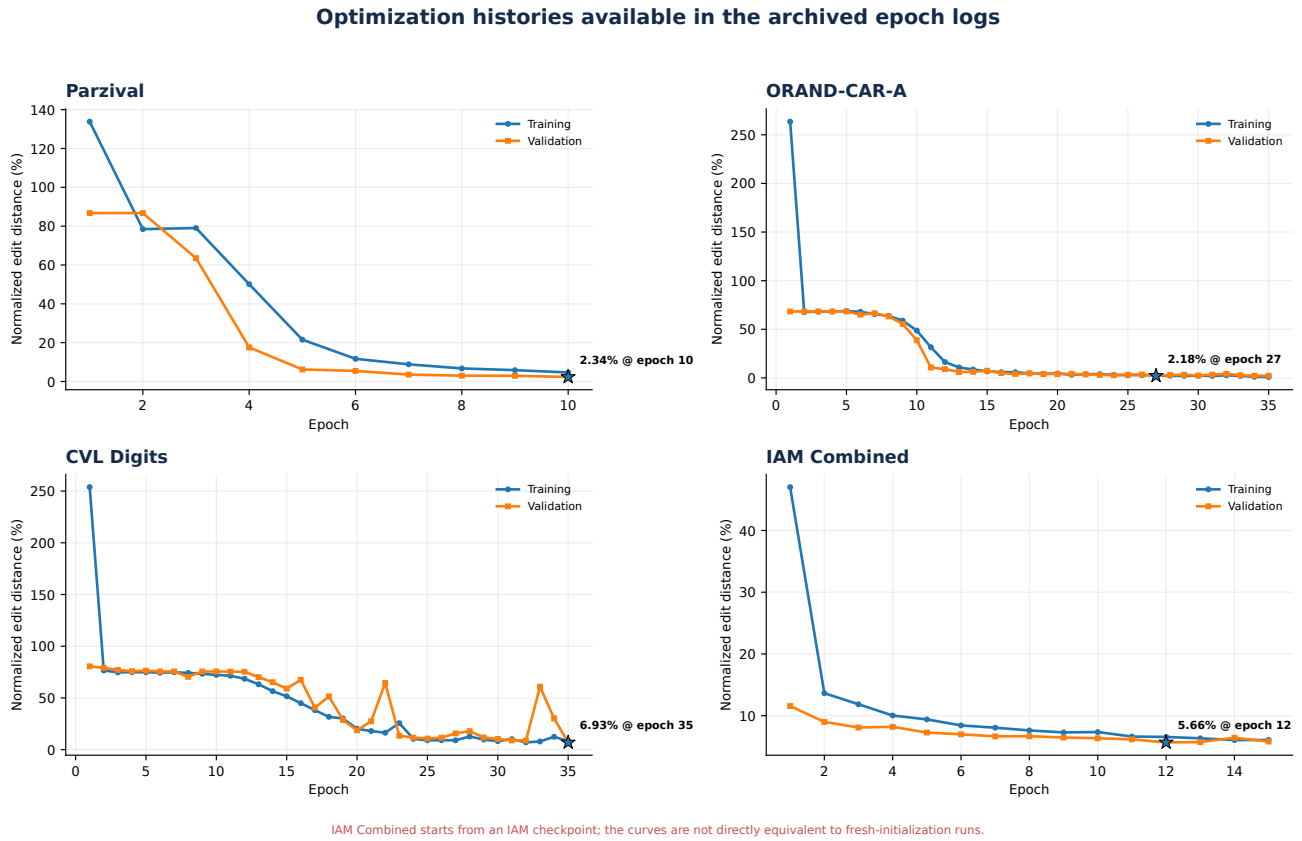


Fig. 8. Training and validation edit-distance trajectories reconstructed from the archived epoch CSV files. Curves are shown for Parzival, ORAND-CAR-A, CVL Digits, and IAM Combined. The star marks the best recorded validation epoch. IAM Combined uses a warm-started checkpoint.

identify which factor is responsible. We therefore report the discrepancy as an observed limitation and do not label it as proven distribution shift without sample-level and writer-level analysis.

D. Parameter Scaling

The output projection changes with the tokenizer vocabulary size. Equation 8 gives 1,566,114 parameters for $C = 14$, 1,576,908 for $C = 56$, and 1,583,847 for $C = 83$. These values are consistent with the stored tensor dimensions for the corresponding checkpoints. The Saint Gall checkpoint is not available for direct tensor verification and is excluded from the parameter plot.

E. Comparison with Published HTR Research

Published HTR results provide context, but a numerical ranking is only meaningful when the recognition unit, dataset split, text normalization, external pretraining, language-model use, and metric aggregation are comparable. For example, the Vertical Attention Network reports paragraph-level recognition, while the present study evaluates extracted line or string samples. TrOCR and DTrOCR use pretrained Transformer components, whereas the present recognizer is a compact CNN-attention-BiLSTM model with dataset-specific training. HTR-VT uses a Vision Transformer encoder and a distinct training recipe. LoGo-HTR adds large-scale self-supervised pretraining on an external handwriting collection [3]–[5], [7], [8].

The published values summarized in Fig. 7 illustrate the range of architectural and data strategies explored in recent HTR research. The 2.75% result reported by Hernandez Diaz *et al.* uses internal and public training data together with an explicit language model; TrOCR relies on pretrained visual and text Transformers; and LoGo-HTR uses large-scale self-supervised pretraining on an external handwriting collection [2], [3], [8]. HTR-VT represents a data-efficient Vision Transformer encoder approach [7]. These systems provide relevant context for the compact parameterization of the present recognizer, but the absence of a held-out IAM test result for IAM Combined prevents a numerical performance ranking. A smaller parameter count should not be interpreted as superior accuracy or faster inference without matched measurements.

VI. Discussion and Limitations

Performance differs across the five datasets. Parzival and ORAND-CAR-A have lower CER than Saint Gall and CVL Digits, and the 43.85-point CVL validation-to-test gap requires sample-level investigation. Aggregate scores cannot establish whether the gap arises from writer composition, string length, character frequency, image quality, or transcript normalization. Because the corpora differ in recognition unit and handwriting domain, their scores are not pooled and do not establish cross-dataset transfer.

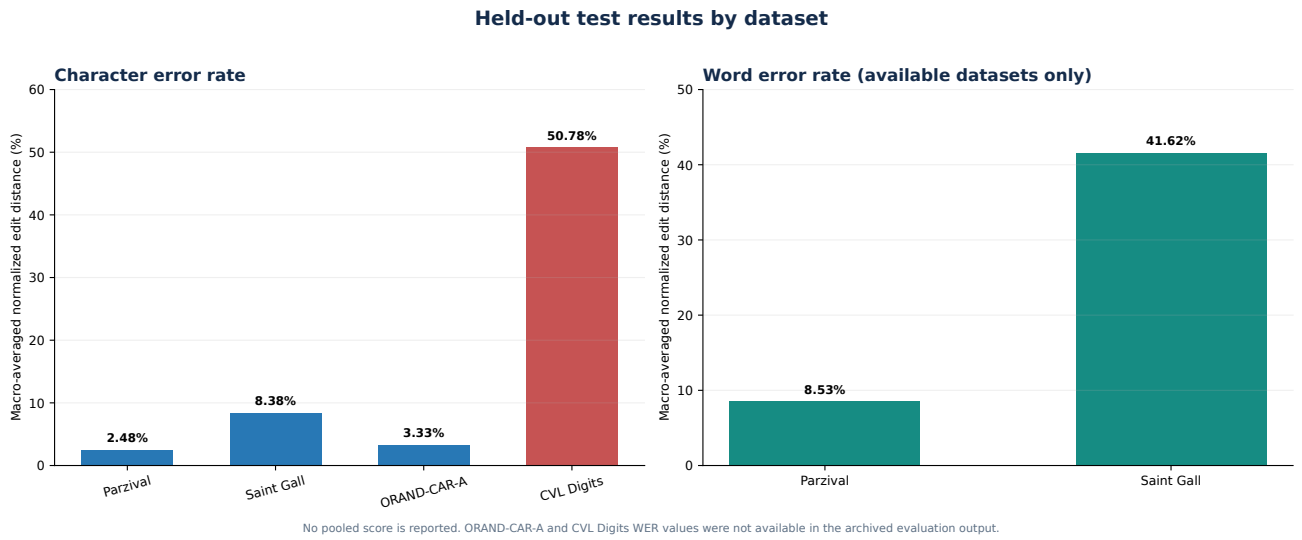


Fig. 9. Held-out test CER for all four evaluated datasets and WER for datasets where the archived notebook reports it. The axes are separated because WER is not available for ORAND-CAR-A and CVL Digits in the cited evaluation output. Values are macro-averaged per-sample normalized edit distances as implemented by the evaluator.

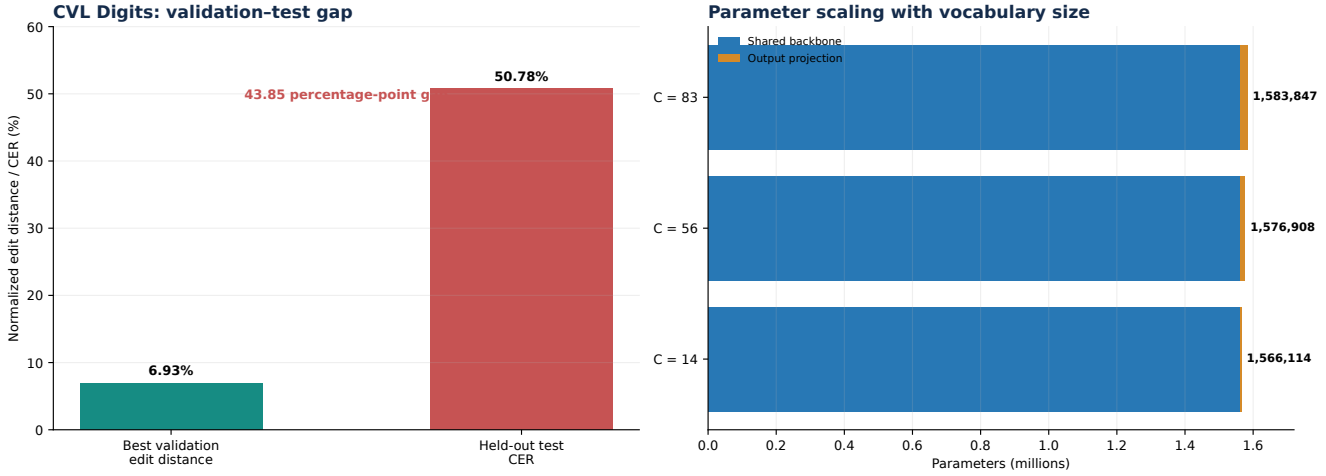


Fig. 10. Two complementary analyses. (a) CVL Digits validation-to-test discrepancy, showing the 6.93% best validation edit distance and 50.78% held-out test CER. The aggregate results do not identify the cause of the gap. (b) Parameter decomposition for the verified output dimensions, showing the shared backbone and dataset-specific projection.

The conditioning operations were not isolated through ablation experiments, and the branch-specific geometric filters may reject genuine handwriting. Reproducibility is incomplete: the Saint Gall notebook test output is available, but its checkpoint and tokenizer are missing; IAM Combined has validation evidence only; and the MLflow database does not track every dataset benchmark. The available artifacts also do not establish calibrated confidence or inference latency. These claims require dedicated sample-level, calibration, and controlled-hardware experiments.

VII. Conclusion

This study documented a compact offline HTR pipeline combining scale-adaptive conditioning, gated convolution, self-attention, BiLSTM, and CTC decoding. Held-out CER was 2.48% on Parzival, 8.38% on Saint Gall, 3.33% on ORAND-CAR-A, and 50.78% on CVL Digits. IAM Combined reached 5.66% validation edit distance, with no independent test result. These limits motivate sample-level error analysis, preprocessing ablations, and repeated evaluations.

Acknowledgment

The authors used ChatGPT (OpenAI) to assist with manuscript organization, language refinement, and preparation of the schematic figures. The authors are responsible for independently verifying the technical descriptions, experimental results, citations, and final manuscript.

REFERENCES

[1] M. Yousef, K. F. Hussain, and U. S. Mohammed, "Recurrence-free unconstrained handwritten text recognition using gated fully convolutional network," in *Proc. 17th Int. Conf. Frontiers in Handwriting Recognition (ICFHR)*, 2020, doi: 10.1109/ICFHR2020.2020.00015.

[2] D. Hernandez Diaz, S. Qin, R. Ingle, Y. Fujii, and A. Bissacco, "Rethinking text line recognition models," arXiv:2104.07787, 2021.

[3] M. Li *et al.*, "TrOCR: Transformer-based optical character recognition with pre-trained models," in *Proc. AAAI*, vol. 37, no. 11, pp. 13094–13102, 2023, doi: 10.1609/aaai.v37i11.26538.

[4] M. Fujitake, "DTrOCR: Decoder-only Transformer for optical character recognition," in *Proc. IEEE/CVF Winter Conf. Applications of Computer Vision (WACV)*, 2024, pp. 8010–8020, doi: 10.1109/WACV57701.2024.00784.

[5] D. Coquenet, C. Chatelain, and T. Paquet, "End-to-end handwritten paragraph text recognition using a vertical attention network," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 508–524, 2023, doi: 10.1109/TPAMI.2022.3144899.

[6] D. Coquenet, C. Chatelain, and T. Paquet, "DAN: A segmentation-free document attention network for handwritten document recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 7, pp. 8227–8243, 2023, doi: 10.1109/TPAMI.2023.3235826.

[7] Y. Li, D. Chen, T. Tang, and X. Shen, "HTR-VT: Handwritten text recognition with vision transformer," *Pattern Recognition*, vol. 158, Art. no. 110967, 2025, doi: 10.1016/j.patcog.2024.110967.

[8] S. Mitra, A. Mondal, and C. V. Jawahar, "Learning beyond labels: Self-supervised handwritten text recognition," in *Proc. IEEE/CVF Winter Conf. Applications of Computer Vision (WACV)*, 2026, pp. 6653–6663, doi: 10.1109/WACV61042.2026.00643.

[9] J. Mas-Candela and J. Calvo-Zaragoza, "Exploring recursive neural networks for compact handwritten text recognition models," *Int. J. Document Anal. Recognit.*, 2024, doi: 10.1007/s10032-024-00481-y.

[10] J. Memon, M. Sami, R. A. Khan, and M. Uddin, "Advancements and challenges in handwritten text recognition: A comprehensive survey," *Journal of Imaging*, vol. 10, no. 1, Art. no. 18, 2024, doi: 10.3390/jimaging10010018.

[11] B. Stutzmann *et al.*, "A survey of historical document image datasets," *Int. J. Document Anal. Recognit.*, 2022, doi: 10.1007/s10032-022-00405-8.

[12] A. Graves, S. Fernandez, F. Gomez, and J. Schmidhuber, "Connectionist temporal classification: Labelling unsegmented sequence data with recurrent neural networks," in *Proc. Int. Conf. Machine Learning (ICML)*, 2006, pp. 369–376.

[13] T. Bluche and R. Messina, "Gated convolutional recurrent neural networks for multilingual handwriting recognition," in *Proc. 14th IAPR Int. Conf. Document Analysis and Recognition (ICDAR)*, 2017, pp. 646–651.

[14] U.-V. Marti and H. Bunke, "The IAM-database: An English sentence database for offline handwriting recognition," *Int. J. Document Anal. Recognit.*, vol. 5, pp. 39–46, 2002.

[15] F. Kleber, S. Fiel, M. Diem, and R. Sablatnig, "CVL-Database: An off-line database for writer retrieval, writer identification and handwriting recognition," in *Proc. ICDAR*, 2013, pp. 560–564.